Main Figures

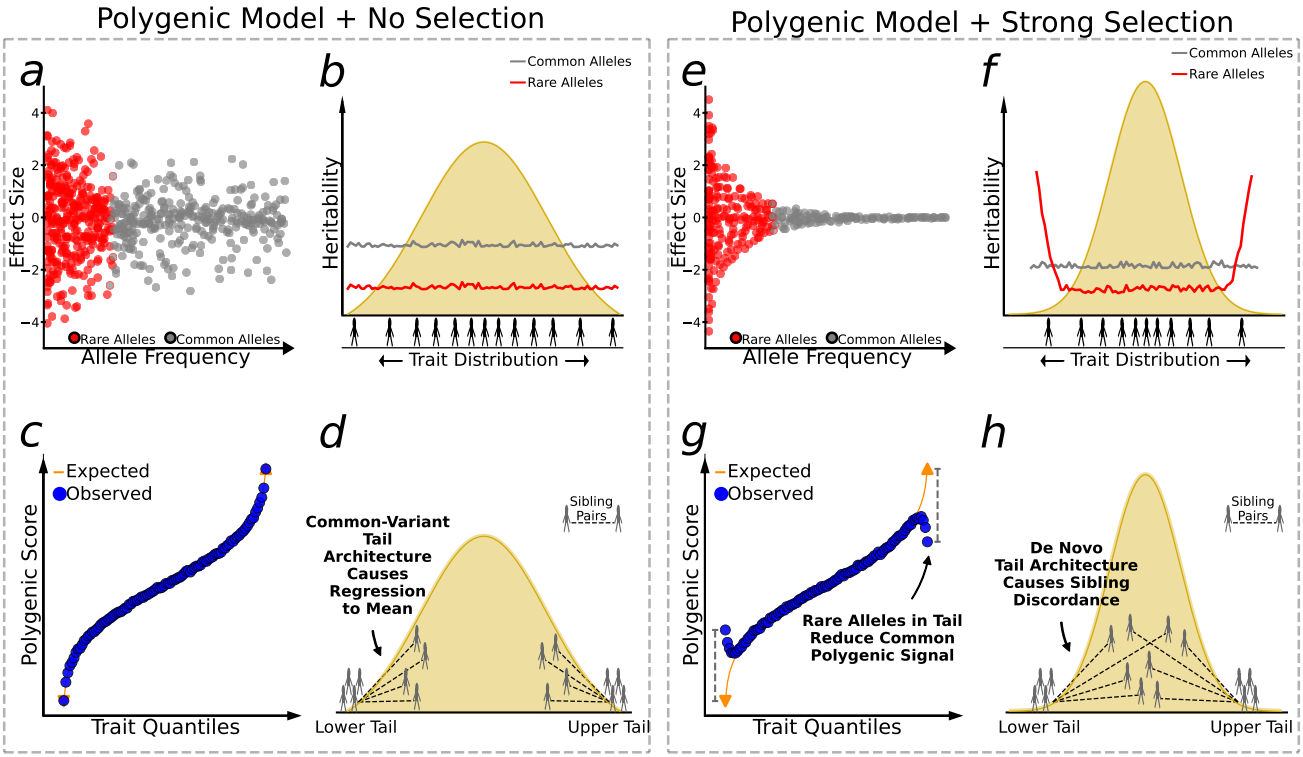


Fig. 1: Two polygenic models of complex trait genetic architecture.

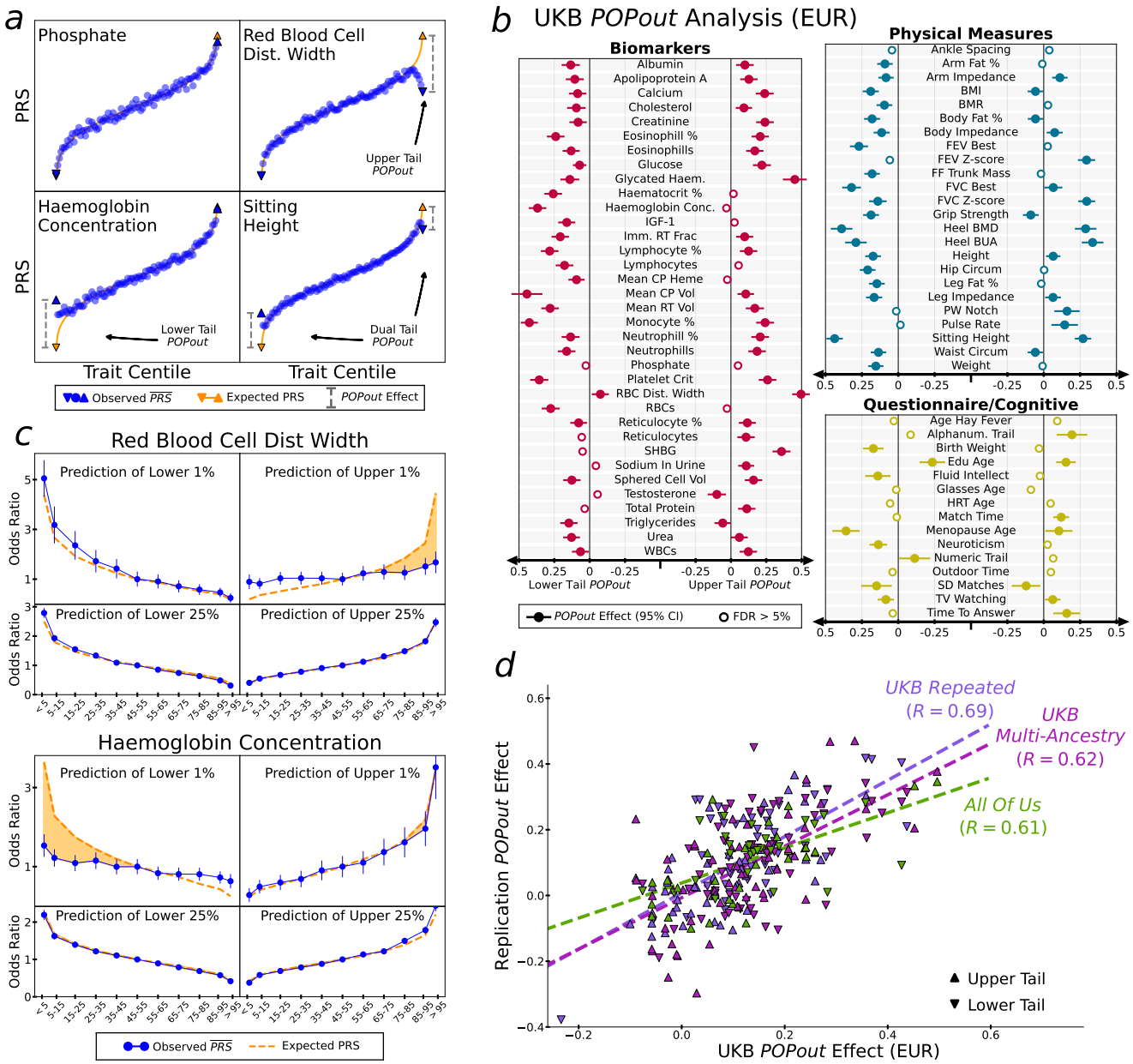


Fig. 2: Population-based POPout analyses.

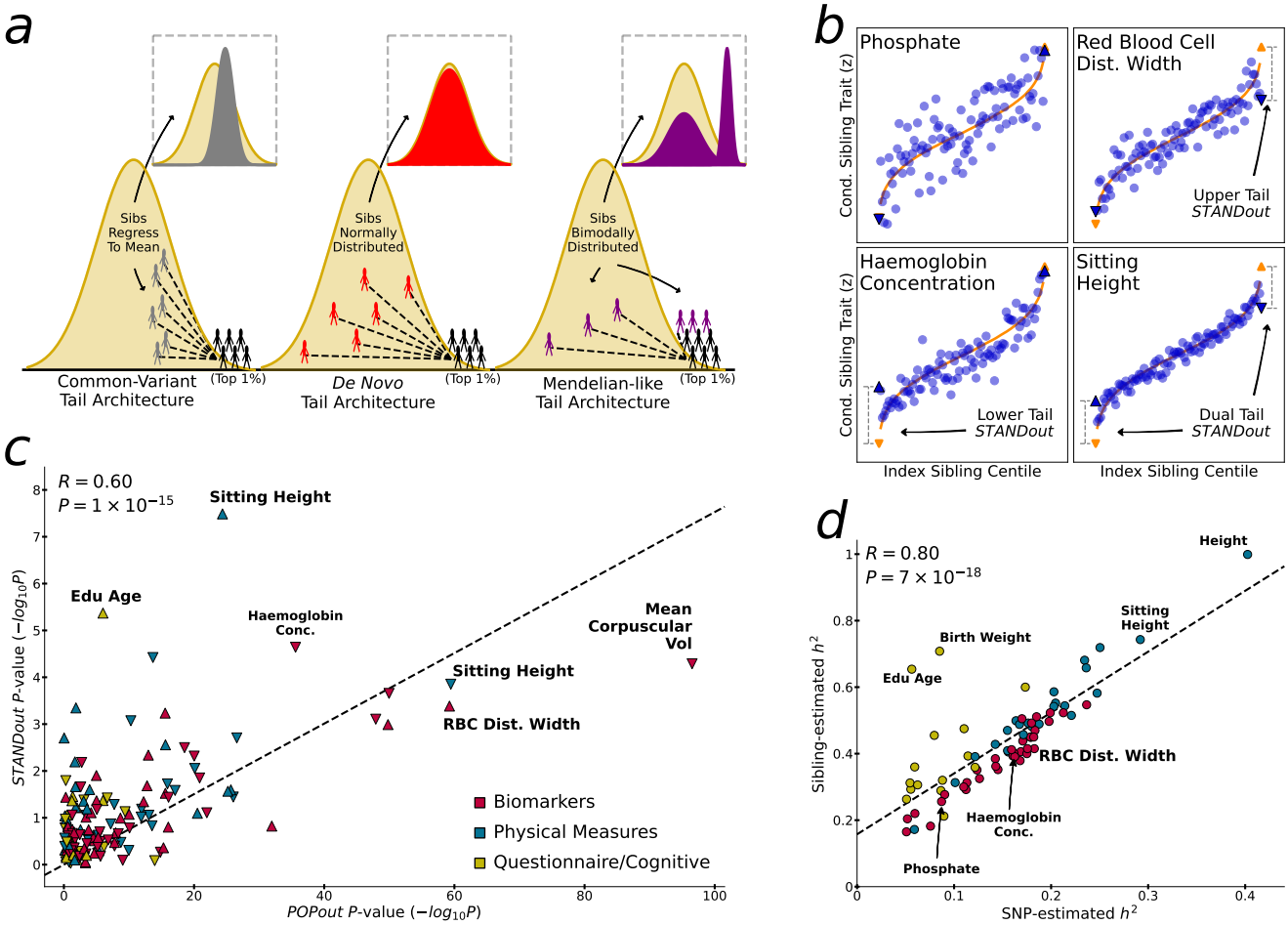


Fig. 3: Tail architecture inferred from siblings.

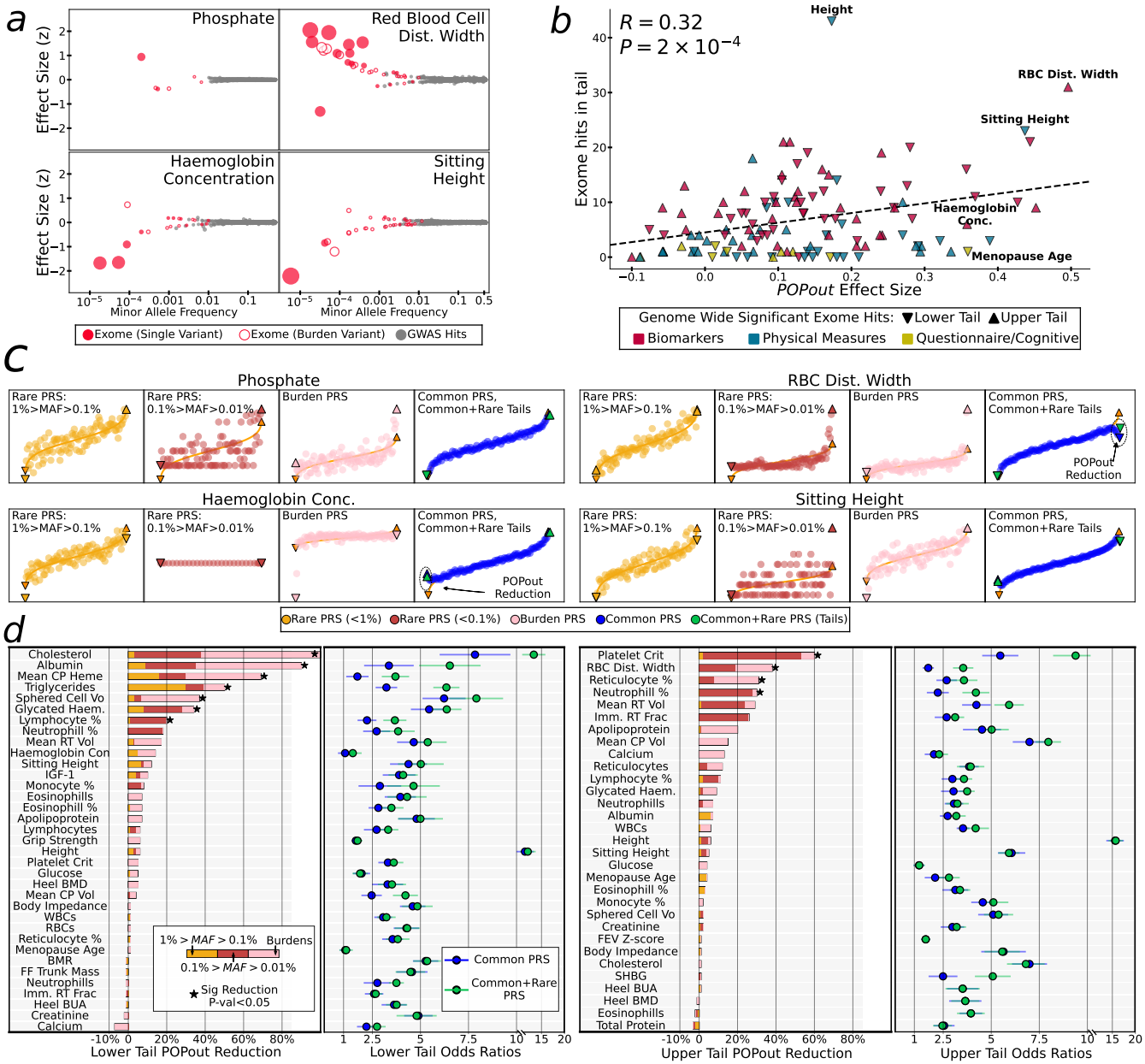


Fig. 4: The role of rare variants in observed POPout effects.

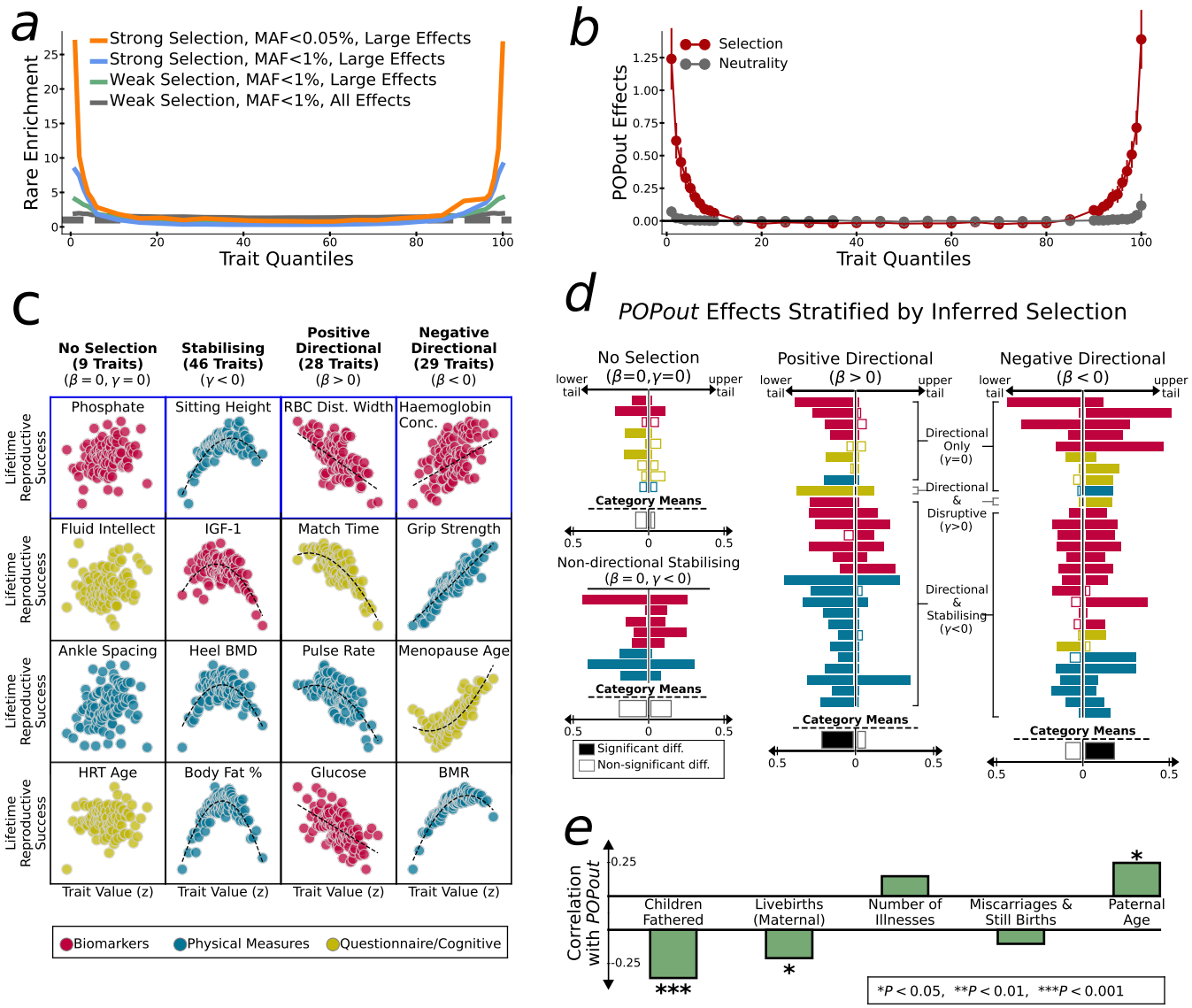
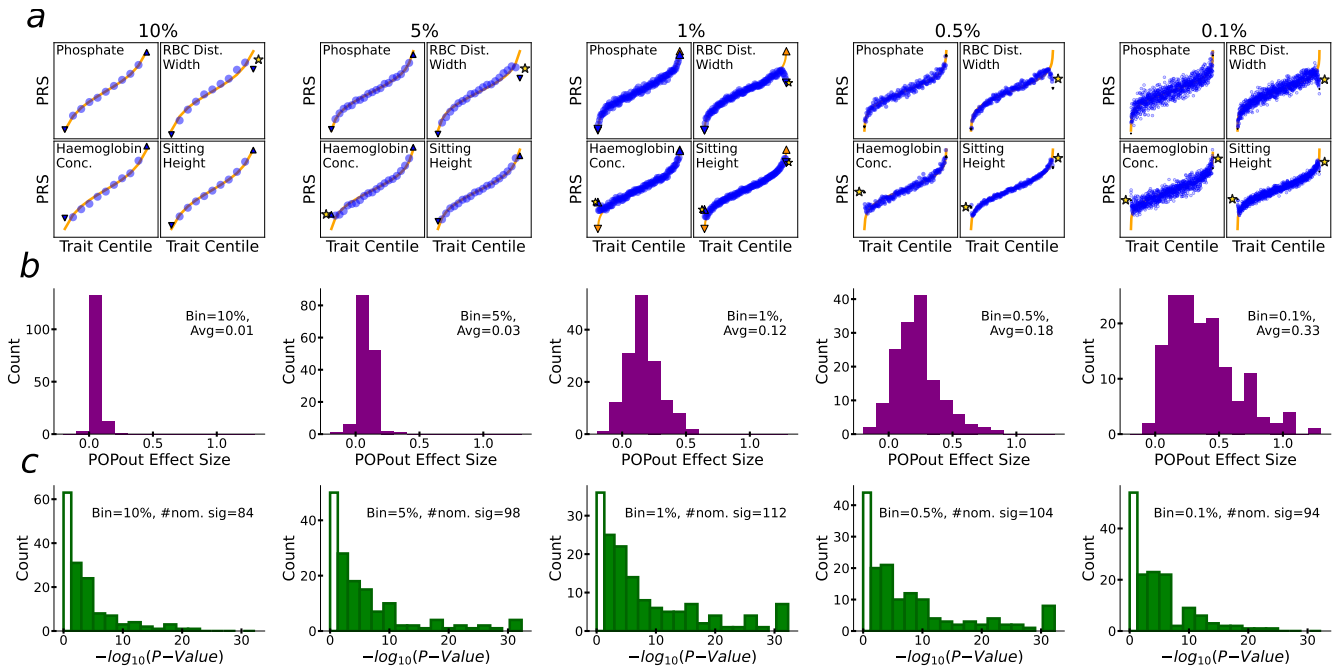
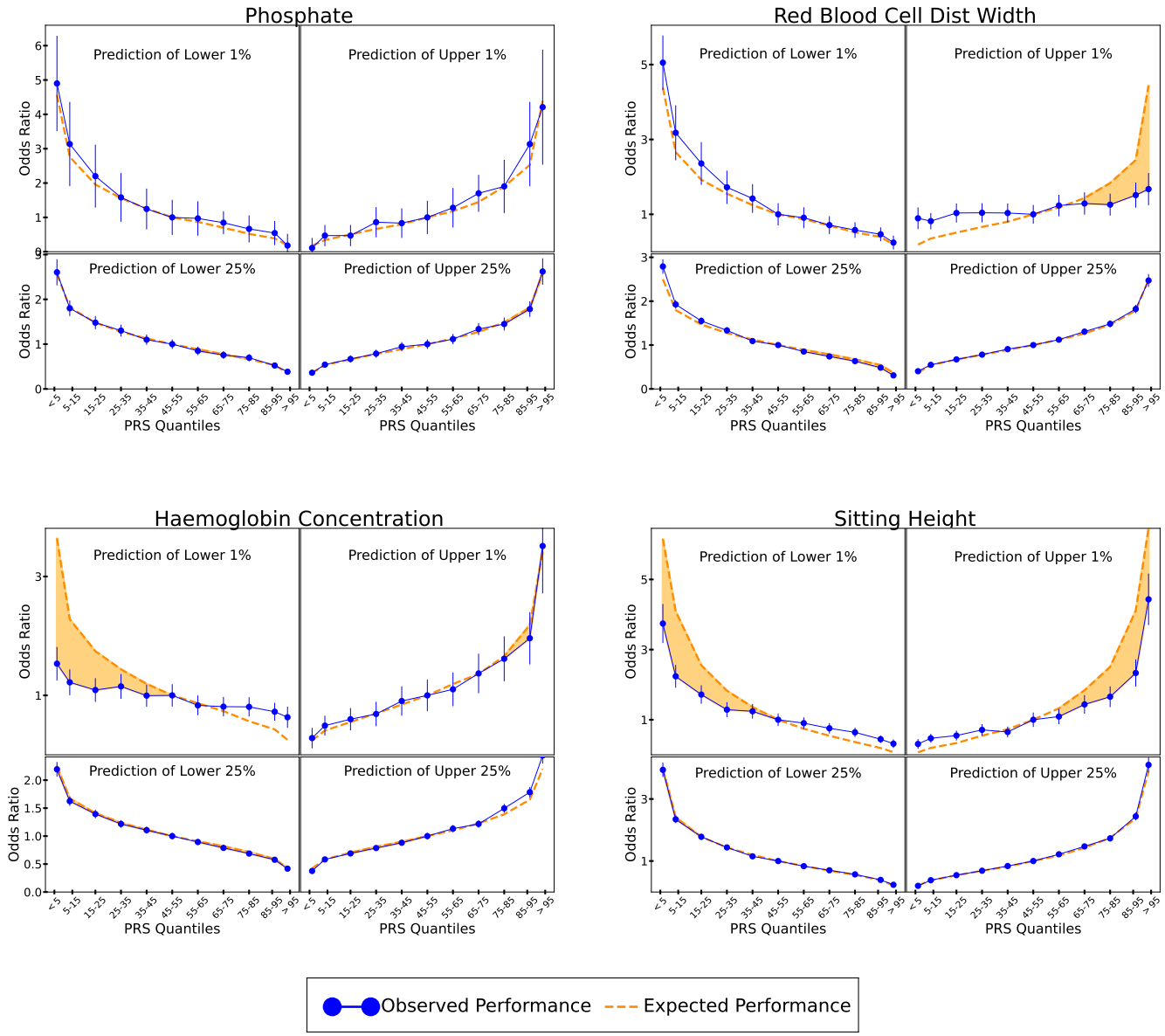


Fig. 5: The role of natural selection in observed POPout effects.

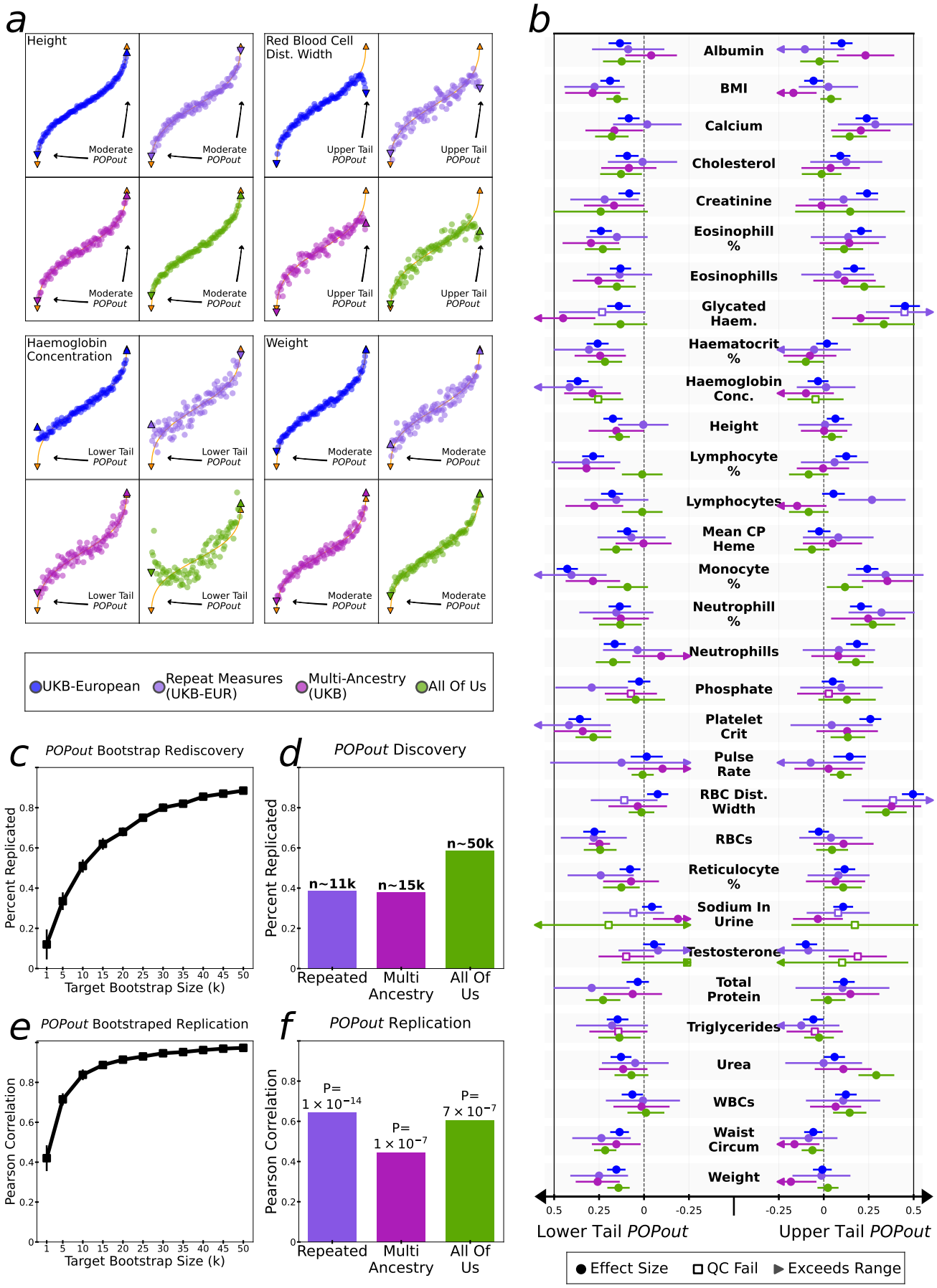
Extended Data



Extended Data Figure 1: Alternative POPout Thresholds (common variants) a, Index trait PRS-on-Phenotype plots for different POPout tail thresholds. Nominally significant POPout effects denoted by "*". For tail sizes greater than 1%, POPout is reduced. At 10%, only RBC Dist. Width shows significance. **b,** POPout effect size distribution at different thresholds: greater thresholds reduce POPout effect. **c,** POPout P-value distribution at different thresholds: The number of significant P-values is maximised at 1% due to a tradeoff between POPout signal (greater for small bins) and statistical power (greater for large bins).



Extended Data Figure 2: Impact on Prediction for Four Traits Extended analysis from Fig1.c, using PRS to predict trait values in the lower/upper 25% and 1% of the distribution.



Extended Data Figure 3: Extended Replication a, PRS quantile (in 1% increments) plotted against mean trait value in quantile for four traits with different types of POPout effects in the primary analyses. **b,** POPout effect sizes in primary and replication cohorts across the 31 overlapping traits in the All of Us cohort. **c,d,** Considering trait tails with significant POPout results in our large European ancestry analysis ('Discovery') as true positives, the percent of tails replicated ($P < 0.05$) in 100 bootstrapped runs of different sizes (**c**) and in the replication datasets (**d**). **e,f,** Replication here is defined here as a significant Pearson correlation across effect sizes in our bootstrapped subsets (**e**) and in the replication datasets (**f**).

Extended Data Figure 4: POPout Summary Table: For all 74 analyzed traits, summary data and PRS-on-Trait, Conditional Sibling, and Selection Inference Plots

Annotated Rare Variants

DRAGEN-POS	rsId	MAF	Type	Genes(s)	Traits with Genome-Wide Sig-Effects	ClinVar Phenotypes
17:16948873	rs34557412	<1%	missense	TNFRSF13B	↓ Platelet Crit,T-Protein,Lymphocytes	Immunodeficiency
9:132989126	rs150813342	<1%	synonymous	GFI1B	↓ Platelet Crit,Eosinophills	Gfi1b-related
9:5073770	rs77375493	<0.01%	non-coding	JAK2, INSL6	↑ RBCDW,Platelet Crit,Neutrophills ↓ Lymphocytes,Monocytes,Cholesterol	Splenomegaly, Jak2-related, Polycythemia, Myelofibrosis
2:224473975	rs527449132	<1%	deletion	CUL3	↑ FF Trunk Mass,BMR	Pseudohypoaldosteronism
22:17107768	rs149771513	<1%	missense	IL17RA	↑ WBCs,Neutrophills	Immunodeficiency
6:49636655	rs375508949	<0.01%	splice-donor	RHAG	↑ Reticulocytes ↓ Cell Vol,Glycated Haem.	Rhag-related, Stomatocytosis
14:23117837	rs147453535	<1%	intronic	CEBPE	↑ Eosinophills	Granule deficiency
22:17105867	rs140221307	<1%	intronic	IL17RA	↓ Monocytes,Eosinophills	Candidiasis, Il17ra-related
17:78138386	rs144120533	<1%	missense	TMC8	↑ Reticulocytes ↓ Glycated Haem.,RBCDW	Epidermodysplasia verruciformis
1:62597921	rs398122988	<0.01%	intronic	DOCK7, ANGPTL3	↓ Cholesterol,Triglycerides	Angptl3-related
20:62475466	rs200383755	<1%	missense	GATA5	↑ FEV Z-score	Gata5-related, Heart defects
11:116830638	rs138326449	<1%	splice-donor	APOC3	↑ Apolipoprotein A,RBCDW ↓ Triglycerides	Heart disease, Apolipoprotein deficiency
7:100620889	rs80338891	<0.01%	missense	TFR2	↑ CP Vol	Hemochromatosis
5:138566714	rs772570880	<0.01%	deletion frameshift	HSPA9	↑ RBCDW	Hspa9-related, Even-plus syndrome
1:1815790	rs141326438	<0.01%	intronic	GNB1	↑ Neutrophills ↓ Lymphocytes,Eosinophills	Intellectual disability, Myelodysplastic syndrome
12:111447491	rs148636776	<0.01%	missense	SH2B3	↑ Platelet Crit	Polycythemia, Myelofibrosis
20:59024148	rs368923302	<0.01%	missense	TUBB1	↓ Platelet Crit	Thrombocytopenia
3:133748533	rs8177318	<0.01%	intronic	TF	↑ RBCDW ↓ CP Vol	Atransferrinemia
2:218135544	rs61733609	<0.01%	missense	CXCR2	↓ Neutrophills	Cxcr2-related
16:88715797	rs202127176	<1%	missense	PIEZO1	↑ Reticulocytes	Piezo1-related
19:12885019	rs397514445	<0.01%	duplication frameshift	KLF1	↑ RBCDW ↓ CP Vol	Anemia, Klf1-related, Fetal henoglobin qtl
1:158643524	rs200830867	<1%	intronic	SPTA1	↓ RT Vol	Spherocytosis, Spta1-related
3:142469495	rs28910273	<1%	missense	ATR	↓ Cell Vol	Atr-related, Telangiectasia
2:218135259	rs55799208	<1%	missense	CXCR2	↑ Lymphocytes ↓ Neutrophills	Cxcr2-related
1:158613814	rs201514157	<0.01%	missense	SPTA1	↓ Cell Vol	Spherocytosis, Spta1-related
9:104858516	rs188308962	<1%	intronic	ABCA1	↓ Apolipoprotein A	Tangier disease
16:88717174	rs763477215	<0.01%	deletion	PIEZO1	↑ Reticulocytes ↓ Glycated Haem.	Piezo1-related, Stomatocytosis, Lymphatic malformation
15:58548387	rs121912502	<1%	missense	LIPC	↑ Apolipoprotein A	Hyperlipidemia
11:116837023	rs199759119	<0.01%	missense	APOA1	↓ Apolipoprotein A	Amyloidosis
2:43838600	rs56204478	<1%	missense	ABCG5	↑ Cholesterol	Abcg5-related, Sitosterolemia
4:154568456	rs6054	<1%	missense	FGB	↑ Cholesterol	Fgb-related, Afibrinogenemia
19:11113620	rs139624145	<0.01%	missense	LDLR	↑ Cholesterol	Dyslipidemia, Hypercholesterolemia
2:21019741	rs41288783	<1%	missense	APOB	↑ Cholesterol	Hypercholesterolemia
9:137233223	rs199690076	<0.01%	missense	SLC34A3	↑ Creatinine	Bone disease, Slc34a3-related
11:10487356	rs117706710	<1%	missense	AMPD3	↑ Glycated Haem.	Ampd3-related
22:36286753	rs138158369	<0.01%	missense	MYH9	↓ Glycated Haem.	Myh9-related, Hearing impairment
16:176987	rs33960522	<0.01%	missense	HBA1	↑ Glycated Haem.	Hba1-related
16:1791324	rs773502836	<0.01%	non-coding	IGFALS	↓ IGF-1	Short stature
17:63917810	rs5388	<1%	missense	GH1	↑ IGF-1	Decreased response to gh
1:21563212	rs786204530	<0.01%	missense	ALPL	↑ Phosphate	Hypophosphatasia, Hypophosphatemia
12:100532530	rs61755050	<1%	intronic	NR1H4	↓ SHBG	Nr1h4-related
19:49861227	rs201872477	<1%	3pUTR	PNKP	↓ T-Protein	Microcephaly
4:87845484	rs115322331	<1%	missense	MEPE	↓ Heel BUA	Mepe-related
11:68348021	rs41494349	<1%	5pUTR	LRP5	↓ Heel BMD	Osteogenesis imperfecta
22:28695868	rs555607708	<1%	deletion frameshift	CHEK2	↑ Menopause Age	Astrocytoma, Osteosarcoma
12:102402247	rs879384585	<1%	3pUTR	IGF1	↓ Height	Growth delay
15:48411104	rs363847	<1%	synonymous	FBN1	↓ Height	Acromicric dysplasia
5:128332993	rs78727187	<1%	missense	FBN2	↑ Height	Ehlers-danlos syndrome

Extended Data Figure 5: Rare Annotations I Rare WGS variants in ClinVar

Annotated Burden Variants

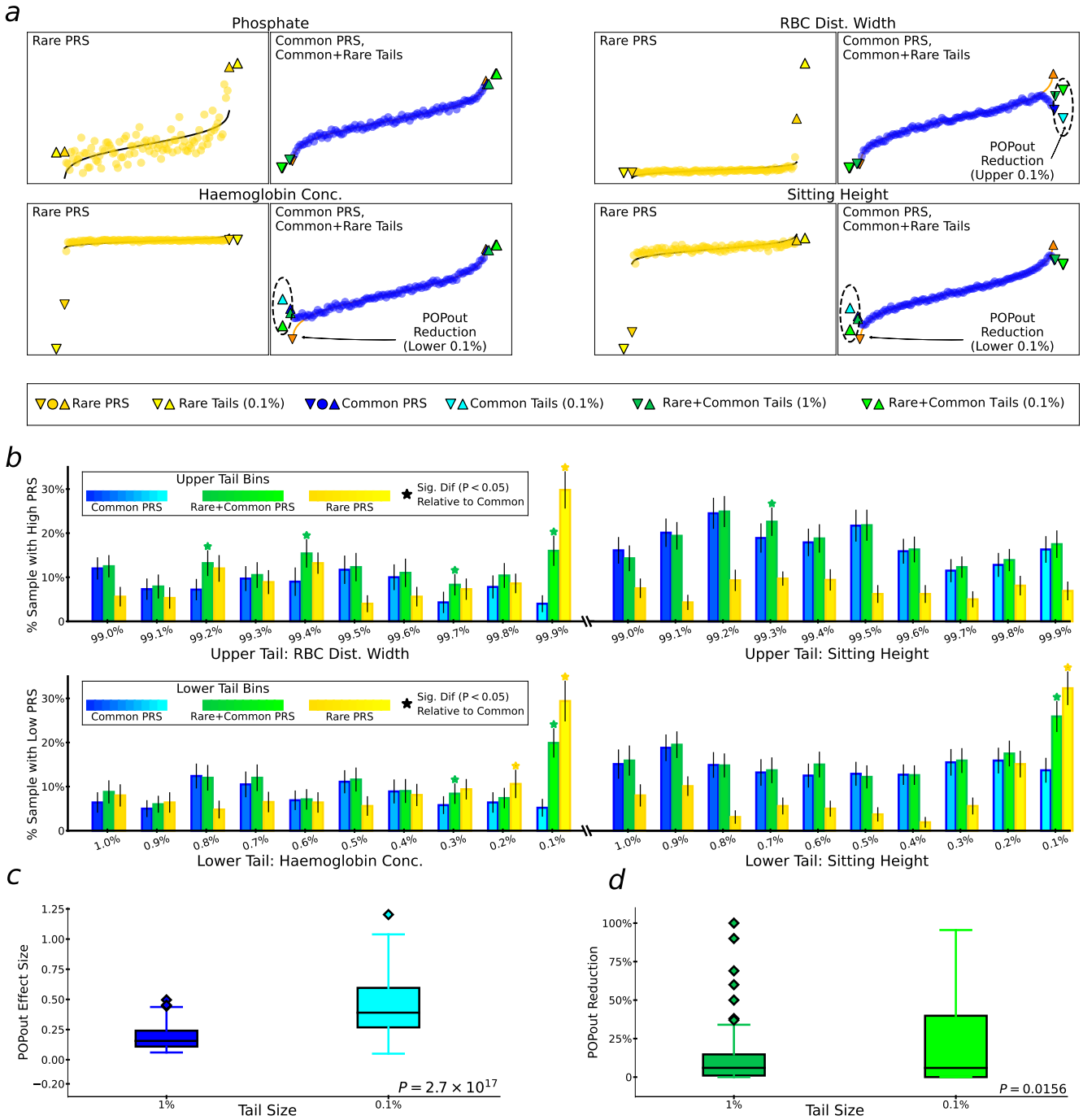
DRAGEN-ID	Genes(s)	Traits with Genome-Wide Sig-Effects	ClinVar Phenotypes
chr15:98649583	IGF1R	↓ Sitting Height,FF Trunk Mass,Height,BMR	Impacts hba1c
chr18:60371355	MC4R	↑ BMR,Ankle Spacing,FF Trunk Mass	Body weight
chr11:5225598	HBB	↑ RBCs,RBCDW,Reticulocytes ↓ CP Vol,CP Heme,Cell Vol,Haemoglobin Conc.	Hemoglobinopathies
chr18:23532206	NPC1	↓ Cell Vol,CP Vol,Monocytes	Intracellular cholesterol trafficking
chr9:35792412	NPR2	↓ Height,Sitting Height	Short stature
chr1:158611274	SPTA1	↑ Reticulocytes,CP Heme ↓ RT Vol,Glycated Haem.,Cell Vol	Elevated retics in hemolysis
chr5:32710664	NPR3	↑ Height,Sitting Height	Height
chr15:88836207	ACAN	↓ Height,Sitting Height	Spine/limb proportion effects
chr17:44250459	SLC4A1	↑ Reticulocytes,CP Heme ↓ Cell Vol,Glycated Haem.	Elevated retics in hemolysis
chr9:6241695	IL33	↓ Eosinophills	Type 2 immunity
chr14:64749312	SPTB	↓ Cell Vol,RT Vol	Elevated retics in hemolysis
chr7:100620858	TFR2	↑ CP Vol,Haemoglobin Conc. ↓ Glycated Haem.,RBCDW	Anemia, Iron metabolism
chr9:104784322	ABCA1	↓ Apolipoprotein A,Cholesterol	Hdl biogenesis
chr2:21001734	APOB	↓ Cholesterol,Triglycerides	Defective apob
chr1:28987430	EPB41	↑ RBCDW,Reticulocytes ↓ Glycated Haem.,Cell Vol	Elevated retics in hemolysis
chr2:168901344	G6PC2	↓ Glucose,Glycated Haem.	Beta-cell glucose handling (fasting glucose)
chr17:7630182	SHBG	↓ SHBG,Testosterone	Testosterone availability via binding
chr22:37066087	TMPRSS6	↑ RBCDW ↓ CP Vol,Haemoglobin Conc.	Iron defect, Microcytosis variability
chr3:196051948	TFRC	↓ CP Vol	Iron uptake, Transferrin receptor
chr20:18510821	SEC23B	↑ RBCDW	Ineffective erythropoiesis
chr19:10350835	TYK2	↑ Lymphocytes ↓ Neutrophills	Cytokine signaling, Allergic disease link
chr20:59019520	TUBB1	↓ Platelet Crit	Platelet formation
chr9:5021988	JAK2	↑ Platelet Crit	Neutrophilia
chr4:105233898	TET2	↑ Monocytes ↓ Eosinophills	Neutrophil counts, Clonal hematopoiesis
chr1:43337853	MPL	↑ Platelet Crit	Thrombopoietin receptor
chr1:36466101	CSF3R	↑ Neutrophills	Mpn, Congenital neutrophilia
chr4:73404338	ALB	↑ Cholesterol ↓ Albumin,Calcium	Major serum protein, Hypoalbuminemia lowers total protein
chr15:58431991	LIPC	↑ Apolipoprotein A	Remodels tg-rich lipoproteins
chr18:49561688	LIPG	↑ Apolipoprotein A	Hdl metabolism, Endothelial lipase
chr16:172952	HBA2	↑ RBCs ↓ CP Vol	Hemoglobinopathies
chr8:19939445	LPL	↓ Apolipoprotein A	Hydrolysis of tg-rich lipoproteins
chr16:56961984	CETP	↑ Apolipoprotein A	Transfers ce/tg between lipoproteins
chr9:137231703	SLC34A3	↑ Creatinine ↓ Phosphate	Napi-iic, Hypophosphatemia
chr11:116790132	APOA5	↓ Apolipoprotein A	Key regulator of plasma tg
chr3:122254194	CASR	↑ Calcium	Secondary effects via pth/calcium
chr7:44513457	NPC1L1	↓ Cholesterol	Cholesterol absorption (ezetimibe target)
chr2:43813117	ABCG5	↑ Cholesterol	Phytosterolemia, Sterol transport
chr1:55039839	PCSK9	↓ Cholesterol	Ldlr degradation
chr19:11089550	LDLR	↑ Cholesterol	Ldl clearance
chr17:19533942	SLC47A1	↑ Creatinine	Mate1
chr6:160217441	SLC22A2	↑ Creatinine	Renal handling
chr7:44145143	GCK	↑ Glycated Haem.	Impacts hba1c, Glycemia driver
chr16:88715607	PIEZO1	↓ Glycated Haem.	Rbc count/volume
chr8:117135328	SLC30A8	↓ Glycated Haem.	Impacts hba1c, Beta-cell function
chr6:131808036	ENPP1	↓ Phosphate	Phosphate handling
chr1:21554084	ALPL	↑ Phosphate	Hypophosphatasia, Alters phosphate
chr19:8364323	ANGPTL4	↓ Triglycerides	Raises tg, Lpl inhibitor
chr19:7117057	INSR	↓ Triglycerides	Impacts hba1c, Insulin resistance
chr11:68312715	LRP5	↓ Heel BMD	Axial skeleton, Skeletal accrual
chr12:48978651	WNT1	↓ Heel BMD	Skeletal development
chr2:219055210	IHH	↓ Height	Indian hedgehog, Skeletal growth
chr7:41964330	GLI3	↑ Height	Hedgehog pathway

Extended Data Figure 6: Rare Annotations II Rare Burden Variants Found in ClinVar WGS variants

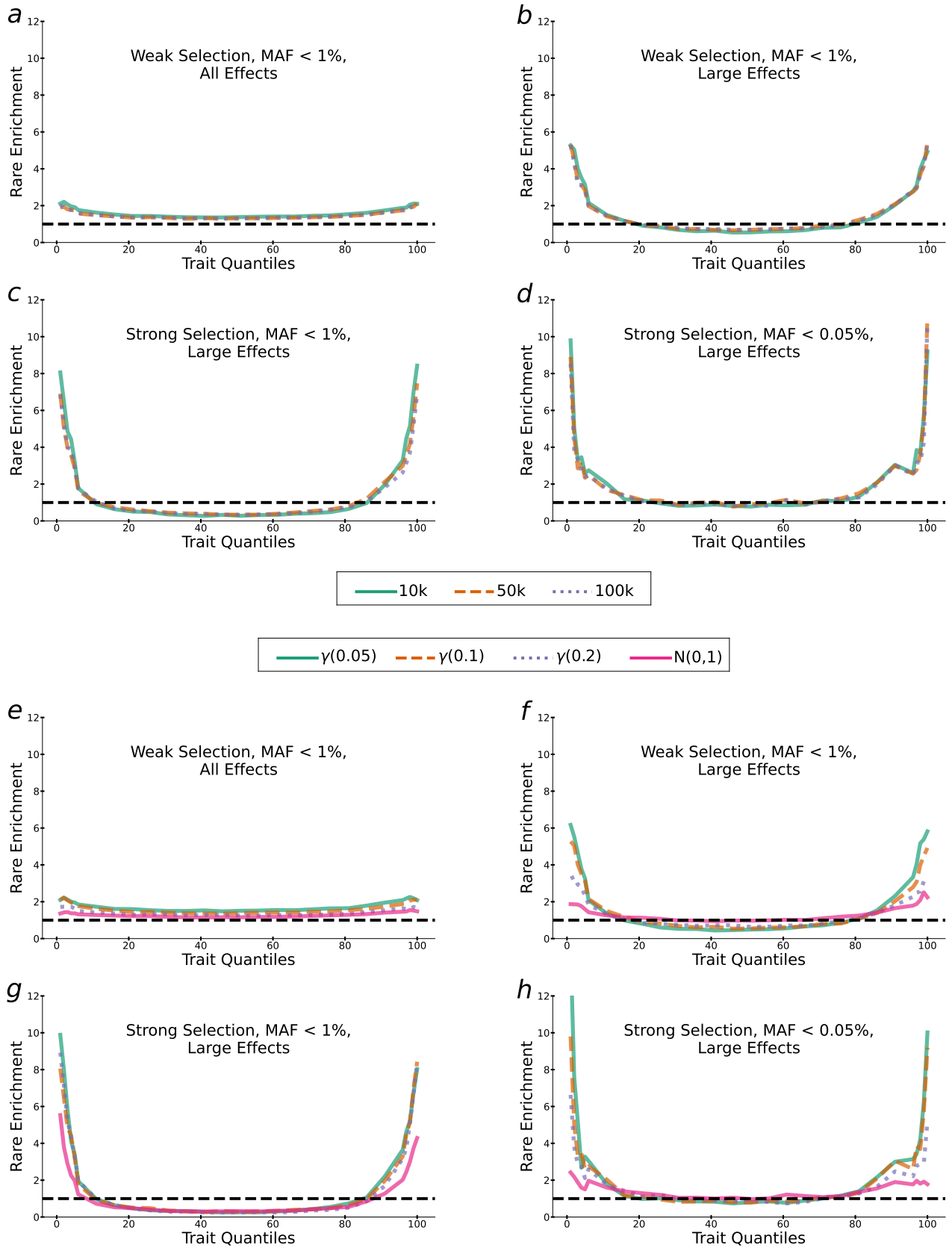
Unannotated Burden Variants

DRAGEN-ID	Genes(s)	Traits with GWS Increasing Alleles	Traits with GWS Descreasing Alleles
chr5:76403546	IQGAP2	Reticulocytes,Reticulocytes	Albumin
chr11:15112472	INSC	Heel BUA,Heel BMD	-
chr2:136114872	CXCR4	Neutrophills,WBCs	-
chr8:85328561	CA1	Reticulocytes,Reticulocytes	RT Vol
chr8:41655730	ANK1	Reticulocytes,Reticulocytes,CP Heme	RT Vol,Cell Vol,Glycated Haem.
chr2:68735201	ARHGAP25	Monocytes	-
chr2:159805052	LY75	Platelet Crit	-
chr3:184372513	THPO	-	Platelet Crit
chr16:1078870	SSTR5	IGF-1	-
chr12:131960624	EP400	-	Grip Strength
chr1:185734781	HMCN1	Sitting Height	-
chr16:67192232	E2F4	CP Vol,RT Vol,Cell Vol	-
chr19:55181755	PTPRH	Glycated Haem.	Platelet Crit
chr3:142449429	ATR	-	CP Vol,Cell Vol
chr5:65151842	ADAMTS6	-	Height
chr19:12938692	CALR	Platelet Crit	-
chr10:52314434	DKK1	-	Sitting Height
chr13:95433834	CLDN10	Creatinine	-
chr17:7173661	ASGR1	-	Cholesterol
chr6:10873998	GCM2	Calcium	-
chr12:460380	B4GALNT3	Height	-
chr8:74985996	CRISPLD1	-	Height
chr12:111418148	SH2B3	Platelet Crit	-
chr15:72162010	GRAMD2	Height,Sitting Height	-
chr19:49513385	FCGRT	-	Albumin,T-Protein
chr7:6024673	EIF2AK1	RT Frac	-
chr22:50525773	TYMP	-	RT Vol,Cell Vol
chr1:176556332	PAPPA2	IGF-1	-
chr3:185645534	IGF2BP2	-	Height
chr22:28742101	HSCB	RBCDW	-
chr7:80646651	CD36	Apolipoprotein A	-
chr4:74097756	CXCL2	-	Neutrophills
chr1:226961386	ADCK3	-	RT Vol,CP Vol
chr18:23453443	RIOK3	RT Frac,Reticulocytes,Reticulocytes	-
chr13:28004055	FLT3	Monocytes	-
chr3:52495414	STAB1	Albumin,Cholesterol,T-Protein	-
chr4:99307843	ADH1B	Apolipoprotein A	-
chr6:43298371	SLC22A7	Creatinine	-
chr1:202729037	KDM5B	-	Grip Strength
chr3:141487084	RASA2	-	Lymphocytes,Lymphocytes
chr2:96761000	CNNM4	Calcium	-
chr6:35214390	SCUBE3	-	Height,Sitting Height,FF Trunk Mass,BMR
chr3:51984082	ACY1	IGF-1	-
chr15:99974402	ADAMTS17	-	Height,Sitting Height
chr11:64207352	FERMT3	-	RT Frac
chr6:33573818	BAK1	Platelet Crit	-
chr16:11965325	TNFRSF17	-	T-Protein
chr3:148991644	GYG1	-	Height
chr19:40800573	EGLN2	Haemoglobin Conc.	-
chr11:26994527	FIBIN	-	Height,Sitting Height
chr1:85580985	CYR61	Body Impedance	-
chr20:37251215	GHRH	-	Height
chr19:1108214	SBNO2	Lymphocytes	-
chr11:75566314	SERPINH1	-	Height
chr17:42900878	G6PC	-	IGF-1
chr7:123115518	SLC13A1	-	Sitting Height
chr15:41183932	EXD1	-	Height
chr15:65150823	CLPX	Creatinine	-
chr20:63659405	RTEL1	Cell Vol,RT Vol	-
chr2:27217564	CAD	CP Vol,RT Vol,Cell Vol	-
chr19:8580898	ADAMTS10	-	Height
chr17:31095310	NF1	SHBG	Height
chr17:4631603	ALOX15	-	Eosinophills,Eosinophills
chr12:48107374	PFKM	-	Glycated Haem.
chr1:158479880	OR10R2	RT Vol	-
chr5:1052362	SLC12A7	-	RBCDW
chr10:45727306	FAM21C	CP Vol	-
chr4:99336754	ADH1C	IGF-1	-
chr4:72283137	ADAMTS3	-	Height
chr3:49674400	APEH	Glycated Haem.	Haemoglobin Conc.,RBCs
chr11:14644078	PDE3B	Apolipoprotein A	Triglycerides
chr16:2027006	SLC9A3R2	-	Haemoglobin Conc.,RBCs
chr16:2429483	CCNF	CP Vol	-
chr1:155195844	THBS3	-	Height
chr17:58270659	MPO	Neutrophills,WBCs,Neutrophills	Monocytes
chr17:74748851	SLC9A3R1	WBCs	-
chr19:45864461	FOXA3	SHBG	-
chr7:45914822	IGFBP3	-	IGF-1
chr2:218134776	CXCR2	-	Neutrophills

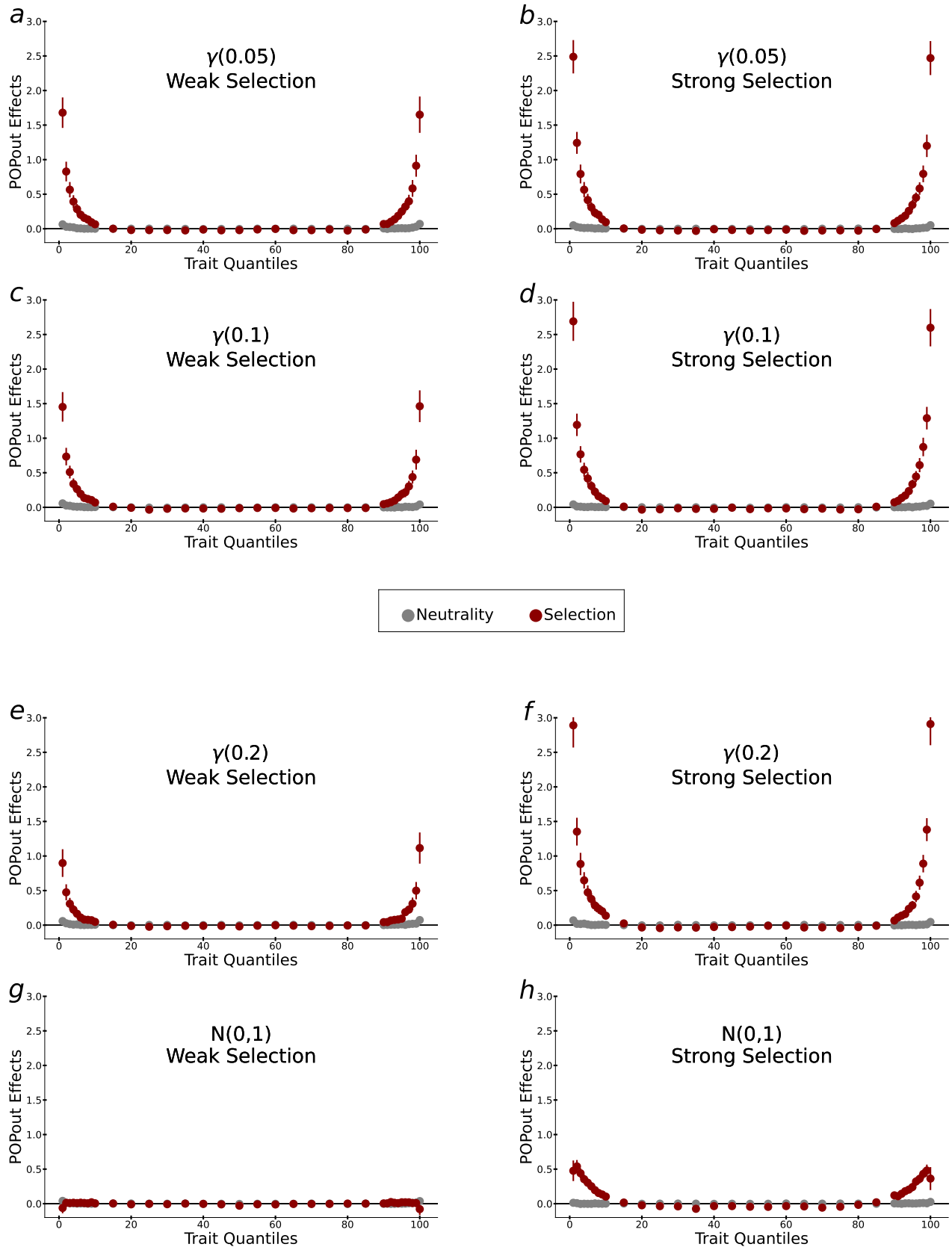
Extended Data Figure 7: Rare Annotations II Rare Burden Variants Genes (No ClinVar Associaion Found)



Extended Data Figure 8: POPout in common, rare, and rare+common PRS models in the standard (1%) and extreme tail (0.1%) a, PRS-on-Phenotype plots that include the rare, common and rare+common PRS means for the top and bottom 1% and the top and bottom 0.1% are shown for our four index traits. **b,** Mechanism behind recovery of tail signal from rare+common PRS is varied. In some cases (Upper Tail: RBC Dist Width) rare variants uniformly "correct" the PRS for individuals in the top 1%, in other cases (Lower Tail: Haemoglobin Conc) recovery is driven primarily through correcting PRS values for a large number the most extreme individuals (bottom 0.1%). **c,** Across 98 trait tails with positive, significant (FDR 5%) common-variant PRS POPout, the average POPout effect is greater when calculated at 0.1% than 1% (mean 0.44 vs 0.19, $P = 2.7 \times 10^{-17}$). **d,** Across 66 trait tails with positive, significant (FDR 5%) common-variant PRS POPout and at least one genome wide statistically significant ($P < 2.18 \times 10^{-}11$) rare-variant identified, POPout recovery is greater at 0.1% than 1% (mean 26% to 13%, $P = 0.0156$).



Extended Data Figure 9: Rare variants are similarly enriched in the trait tails across longer durations of stabilising selection. Enrichment of rare alleles in simulations under weak and strong selection scenarios, stratified by rare variant classification and effect size. For each stratification (A-D), rare variant enrichment is shown across trait quantiles for different durations of stabilising selection (i.e. 10k (green), 50k (orange), and 100k (purple) generations of selection). We observe no major differences in rare variant enrichments between timepoints. The dotted horizontal line represents enrichment under neutrality. For each stratification (E-H), rare variant enrichment is shown across trait quantiles for different type of mutation distributions: gamma (mean=0.005, shape= 0.186) in green, gamma (mean=0.01, shape= 0.186) in orange, gamma (mean=0.02, shape= 0.186) in purple, and Gaussian N(0,1) in pink. For each stratification, we observe rare variant enrichment in the tails after 1N generations of stabilising selection with slight differences in the degree of enrichment depending on the distribution. All three heavy-tailed gamma distributions produced greater enrichment of rare variants in the tails than the gaussian distribution. The dotted horizontal line represents enrichment under neutrality.



Extended Data Figure 10: POPout effects are observed under different mutation distributions. POPout effects are shown across trait quantiles averaged over 100 simulations (mean and CIs shown) under neutrality (in grey) and (weak/strong) selection (in red). Results are visualised stratified by mutation distribution: gamma (mean=0.005, shape= 0.186) in A/B, gamma (mean=0.01, shape= 0.186) in C/D, gamma (mean=0.02, shape= 0.186) in E/F, and Gaussian N(0,1) in G/H. Selection was introduced for 1N generations. We observe strong POPout effects in the tail across different Gamma distributions and strengths of selection. For Gaussian effect sizes, we only observe POPout effects when the simulated trait is subject to strong selection.

Supplemental Figures

Trait Name [UKB ID]	Model Selection			Sanjak et al					
	Top Model	Linear Parameter (β)	Quadratic Parameter (γ)	Aggregate Result		Males		Females	
				β	γ	β	γ	β	γ
Traits Without Sex Differences									
Body Mass Index (bmi) (21001)	$Y \sim X_o + X + X^2$	0.034 (positive)	-0.025 (stabilising)	positive	stabilising	0.072*	-0.0327*	0.0601*	-0.022*
Basal Metabolic Rate (23105)	$Y \sim X_o + X + X^2$	0.042 (positive)	-0.023 (stabilising)	positive	stabilising	0.0747*	-0.03*	0.0405*	-0.019
Birth Weight (20022)	$Y \sim X_o + X$	0.011 (positive)	0 (neutral)	positive	neutral	0.0208*	-0.0016	0.0047	-0.005
Body Fat Percentage (23099)	$Y \sim X_o + X^2$	0 (neutral)	-0.025 (stabilising)	positive	stabilising	0.0198*	-0.0273*	0.0188*	-0.016*
FEV Best (20150)	$Y \sim X_o + X + X^2$	0.023 (positive)	-0.011 (stabilising)	positive	stabilising	0.0335*	-0.0109*	0.0126	-0.004*
FVC Best (20151)	$Y \sim X_o + X + X^2$	0.026 (positive)	-0.013 (stabilising)	positive	stabilising	0.0323*	-0.0141*	0.0165*	-0.007*
Match Time (20023)	$Y \sim X_o + X + X^2$	-0.026 (negative)	-0.007 (stabilising)	negative	stabilising	-0.0434*	-0.01*	-0.0101*	-0.0002
Pulse Rate (4194)	$Y \sim X_o + X + X^2$	-0.031 (negative)	-0.011 (stabilising)	negative	stabilising	-0.0316*	-0.0062*	-0.0136*	-0.0053*
Waist Circumference (48)	$Y \sim X_o + X + X^2$	0.032 (positive)	-0.027 (stabilising)	positive	stabilising	0.0459*	-0.0331*	0.0649*	-0.028*
Weight (21002)	$Y \sim X_o + X + X^2$	0.035 (positive)	-0.028 (stabilising)	positive	stabilising	0.0761*	-0.0343*	0.0439*	-0.0213*
Traits With Sex Differences									
Edu Age (845)	$Y \sim X_o + X + X^2$	-0.044 (negative)	0.003 (disruptive)	positive, negative	disruptive	0.0086	0.0313*	-0.0612*	0.034*
Fluid Intelligence Score (20016)	$Y \sim X_o$	0 (neutral)	0 (neutral)	neutral	neutral	0.0139	0.0009	-0.0473	0.02
Hand Grip Strength (left) (46)	$Y \sim X_o + X$	0.045 (positive)	0 (neutral)	positive	neutral	0.0728*	-0.0023	0.0124*	0.003
Heel Bone Mineral Density (3148)	$Y \sim X_o + X^2$	0 (neutral)	-0.014 (stabilising)	positive, negative	stabilising	0.0325*	-0.0094*	-0.0105	-0.01*
Standing Height (50)	$Y \sim X_o + X^2$	0 (neutral)	-0.019 (stabilising)	positive, negative	stabilising	0.0229*	-0.0152*	-0.0283*	-0.0188*
Neuroticism Score (20127)	$Y \sim X_o + X + X^2$	-0.02 (negative)	-0.011 (stabilising)	negative, positive	stabilising	-0.0318*	-0.006	-0.0003	-0.0061*

Supplemental Figure 1: Replication of Sanjak[?] lifetime reproductive success results. Comparison of results among overlapping traits (without sex differences) shows replication (i.e. statistically significant parameters included in our model) for 19 of 20 parameters. The only exception is *Body Fat*, in which Sanjak et al[?] reported a positive stabilising result in males and females, while we selected a neutral stabilising model in the pooled sample.

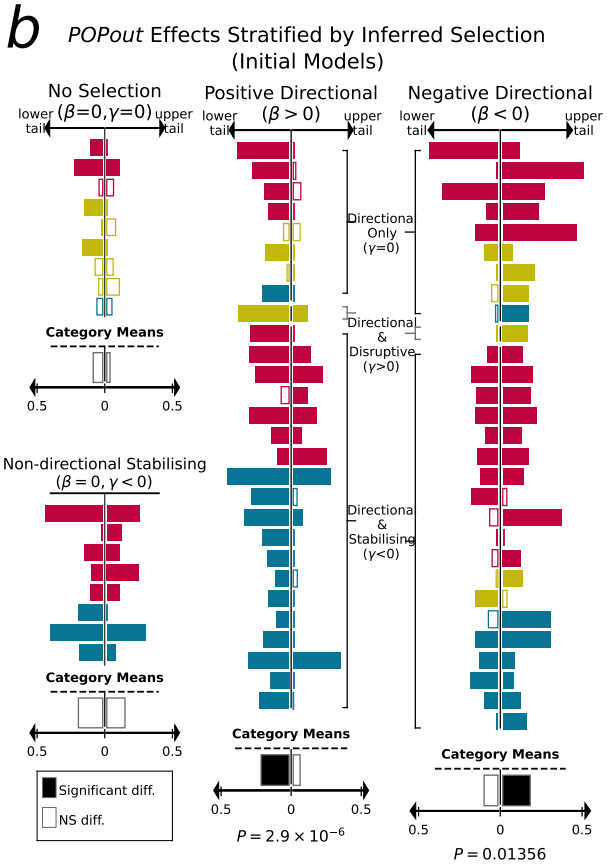
a

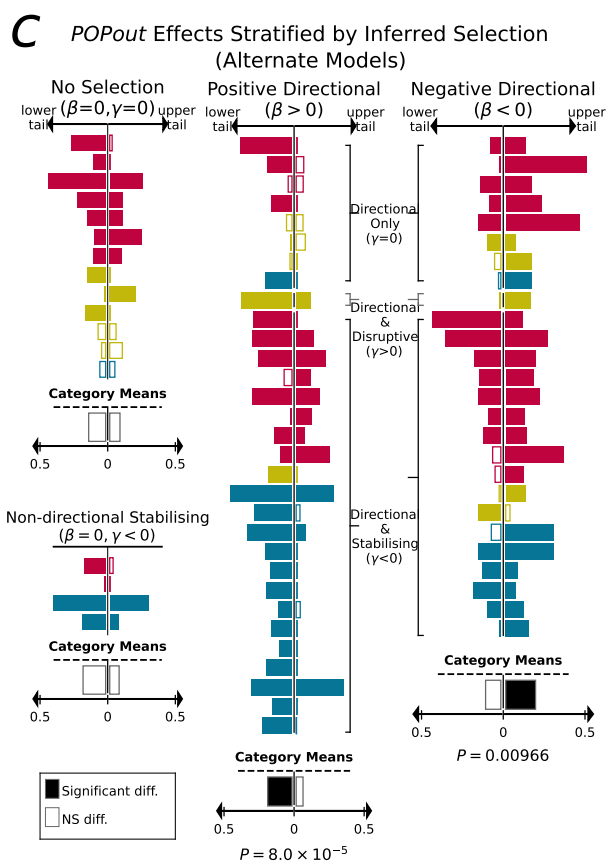
Nested Alternatives

Trait	Top Model			Alternate Model	
	Model	Selective Pressure	Model Confidence (LRT P-Value)	Model	Selective Pressure
Albumin	$Y \sim X_o - X^2$	stabilising	>0.1	$Y\ X_o$	neutral
Alphanum. Trail	$Y \sim X_o - X$	negative	>0.05	$Y\ X_o$	neutral
Calcium	$Y \sim X_o - X^2$	stabilising	>0.05	$Y\ X_o$	neutral
Cholesterol	$Y \sim X_o - X^2$	stabilising	>0.1	$Y\ X_o$	neutral
Haematocrit %	$Y \sim X_o + X$	positive	>0.1	$Y\ X_o$	neutral
IGF-1	$Y \sim X_o - X - X^2$	neg,stabilising	>0.05	$Y\ X_o - X^2$	stabilising
Monocyte %	$Y \sim X_o - X^2$	stabilising	>0.1	$Y\ X_o$	neutral
Sphered Cell Vol	$Y \sim X_o - X - X^2$	neg,stabilising	>0.1	$Y\ X_o - X$	negative
Testosterone	$Y \sim X_o - X - X^2$	neg,stabilising	>0.05	$Y\ X_o - X^2$	stabilising
WBCs	$Y \sim X_o - X - X^2$	neg,stabilising	>0.1	$Y\ X_o - X$	negative

Complex Alternatives

Trait	Top Model			Alternate Model	
	Model	Selective Pressure	ΔBIC	Model	Selective Pressure
Birth Weight	$Y \sim X_o + X$	positive	0.4	$Y\ X_o + X - X^2$	pos,stabilising
Body Fat %	$Y \sim X_o - X^2$	stabilising	0.2	$Y\ X_o + X - X^2$	pos,stabilising
Mean CP Vol	$Y \sim X_o - X$	negative	1.3	$Y\ X_o - X - X^2$	neg,stabilising
Numeric Trail	$Y \sim X_o$	neutral	1.1	$Y\ X_o + X$	positive
Phosphate	$Y \sim X_o$	neutral	0.3	$Y\ X_o + X$	positive
Platelet Crit	$Y \sim X_o - X$	negative	0.6	$Y\ X_o - X - X^2$	neg,stabilising
Sodium In Urine	$Y \sim X_o - X^2$	stabilising	0.4	$Y\ X_o + X - X^2$	pos,stabilising





Supplemental Figure 2: BIC model selection sensitivity analysis. a, For 17 of 74 traits, the top model selected using BIC was not significantly better fitting than the second best model. **b,** Association between BIC selected models of selection and POPout effects (top models). **c,** Association between BIC selected models of selection and POPout effects after instead using the second best model for the 17 traits in which the top model was not significantly better fitting than the second best model.